An exact factor-seven approximation for planar domination in constant locality

AI-generated write-up

23 October 2026

Abstract

We prove that minimum dominating set in finite simple planar graphs admits a deterministic constant-round LOCAL algorithm with approximation ratio exactly 7. The constant is independent of the number of vertices and the maximum degree; the algorithm needs neither of these parameters nor a planar embedding. Together with the known lower bound excluding every ratio $7-\varepsilon$, this shows that the optimal constant is attained. The proof combines a two-threshold reduction of hard common neighborhoods, exact maximal packing with bounded constraint size, a selection threshold that retains quantitative slack, and a planar face-incidence count. The comparison is with the integral domination number. We give the complete algorithm and all the combinatorial and numerical arguments needed for its guarantee, including a finite construction of its constant-round packing routine.

1 The question and the result

A set D of vertices of a graph is *dominating* if every vertex belongs to D or has a neighbor in D. The minimum cardinality of such a set is the *domination number* $\gamma(G)$. We ask how well a distributed algorithm can approximate $\gamma(G)$ when every vertex must decide whether to join the output after a constant number of communication rounds.

Hilke, Lenzen, and Suomela proved that, for every fixed $\varepsilon > 0$, no deterministic constant-round algorithm achieves approximation ratio $7-\varepsilon$ on planar graphs [5]. Recent work of Amiri and Siebertz [9] and Picavet [10] gives ratio $7+\varepsilon$ for every fixed $\varepsilon > 0$. These bounds identify the infimum of the achievable ratios as 7, but leave the endpoint question:

Is there one constant-round algorithm with ratio exactly 7?

The answer is affirmative.

Theorem 1.1. *There are a deterministic LOCAL algorithm and an absolute constant r such that, on every finite simple planar graph G, the algorithm terminates within r rounds and outputs a dominating set O with*

$$|O| \le 7\gamma(G).$$

The round bound is independent of $|V(G)|$, the maximum degree, the identifier assignment, and the port numbering. The algorithm is not given a planar embedding, $|V(G)|$, or the maximum degree.

The guarantee has no additive error and no minimum-order restriction. In particular, disconnected graphs, isolated vertices, and the empty graph are included. The constants in the construction are deliberately generous; no useful practical running time is claimed.

1.1 Prior work and the techniques used here

In the deterministic LOCAL model [1, 2], the established constant-round approximation ratios for planar minimum dominating set have progressed from 130 [3], through 52 [4] and $11 + \varepsilon$ [6, 7], to $8 + \varepsilon$ [8] and, most recently, $7 + \varepsilon$ [9, 10]. Hilke, Lenzen, and Suomela [5] proved that every ratio $7 - \varepsilon$ is impossible in constant rounds. Amiri and Siebertz [9] also give a path-power lower-bound construction, including a formulation with known network size and identifiers drawn from $\{1, \ldots, n\}$. These bounds identify seven as the infimum; attaining that infimum requires an additional argument. In other local optimization problems, an endpoint approached by all $\alpha + \varepsilon$ guarantees can remain unattainable [19].

The two recent $7+\varepsilon$ algorithms have different scope and techniques. Amiri and Siebertz [9] use clustering, averaging, and LP derandomization on proper minor-closed classes; their planar guarantee is against the fractional domination optimum $\gamma_f(G)$. Picavet [10] works under bounds on maximum average degree and on the density of depth-one shallow minors, with a guarantee against the integral optimum $\gamma(G)$. Maximum average degree means the largest average degree of a nonempty subgraph. A depth-one shallow minor is obtained by contracting disjoint connected subgraphs of radius at most one and then deleting vertices and edges; for a nonempty minor, its density is its number of edges divided by its number of vertices. The exact-seven result here also compares to $\gamma(G)$. It does not establish a factor seven against $\gamma_f(G)$. Here $\gamma_f(G)$ is the minimum of $\sum_{v \in V} x_v$ over nonnegative real weights satisfying $\sum_{v \in N[w]} x_v \geq 1$ for every vertex w, where $N[w]$ is the vertex w together with its neighbors; integral dominating sets are feasible zero–one solutions, so $\gamma_f(G) \leq \gamma(G)$.

The present construction starts from Picavet's hard-neighborhood hypergraph and sunflower framework [10, 20], which in turn builds on the degree-reduction and LP approach of Heydt and coauthors [6, 7]. We also use the threshold/layer-cake viewpoint in Picavet's rounding analysis. The endpoint proof separates two structural thresholds, computes an exact maximal fractional packing without a bound on variable frequency, and charges missed hard neighborhoods to quantitative rounding slack and planar Euler deficits. Every lemma used for these steps is proved here.

Density-based domination rounding predates these local algorithms. The arboricity a of a graph is the minimum number of forests whose edge sets cover all of its edges. Bansal and Umboh [11] give a $3a$ approximation for graphs of arboricity at most a positive bound a, and Dvořák [12] improves the relevant rounding bound to $2a + 1$. Dory, Ghaffari, and Ilchi [13] obtain $(2a+1)(1+\varepsilon)$ in degree-dependent distributed time. Earlier constant-round approaches use local approximate covering-LP solvers of Kuhn, Moscibroda, and Wattenhofer [14, 15] after reducing the incidence parameters. Our packing lemma instead needs exact maximality, not an optimal packing objective. Exact fractional-packing algorithms have precedents in Åstrand and Suomela [16]; the bounded-row-arity, finite-capacity formulation used here is proved directly.

The endpoint does not follow by substituting $\varepsilon = 0$ in a $(7+\varepsilon)$-approximation: the number of rounds of that family may diverge as ε tends to zero. The proof below instead fixes all parameters once and accounts for the apparent approximation losses using nonnegative slack in the same solution.

1.2 Proof outline

First, we mark single vertices and pairs whose common neighbors cannot be dominated cheaply from outside the marked set. A reduction with two distinct thresholds produces a hypergraph $\mathcal{H}$ of rank at most two and bounded maximum degree. Every original hard set contains a retained hyperedge. After removing the common neighbors of retained hyperedges from the request set, each vertex has only a bounded number of residual neighbors.

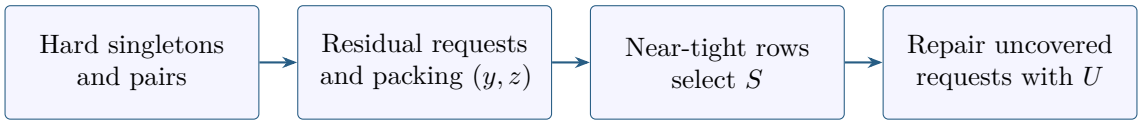


Figure 1: The algorithm. All four stages use a fixed-radius neighborhood. The marked hyperedges help dominate the removed requests; the repair set U handles the residual requests not already dominated by S.

Next, we compute a maximal fractional packing with variables y_w for residual requests and z_e for retained hyperedges. Its rows have a fixed maximum size, even though a variable can occur in arbitrarily many rows. We prove that fixed rational capacities and bounded row size allow an exact constant-round algorithm for this packing.

Finally, we select vertices whose packing rows are sufficiently close to tight, and add the residual vertices still undominated. The planar edge bound yields a quantitative rounding inequality. Comparison with an arbitrary integral dominating set produces a common non-negative budget. A face count and star contractions show that hyperedges missed by the comparison set consume only a small fraction of that budget. The remaining slack gives the exact factor 7.

2 Model, notation, and elementary planar facts

We use the standard synchronous deterministic LOCAL model [1, 2]. Vertices are processors with distinct positive integer identifiers. An edge permits a message in each direction in each round; messages and local computations have no size or time bound. Incident edges may have arbitrary local port numbers. The algorithm must be effective: every local step is an ordinary terminating computation. Exact operations on rational numbers and exhaustive searches over a finite neighborhood are permitted. In a constant number of rounds a vertex can gather a fixed-radius neighborhood, together with identifiers and the adjacency information needed below. The graph is promised to be planar; the algorithm does not test this promise.

All graphs are finite, undirected, and simple. Unless specified otherwise, adjacency and neighborhoods refer to the input graph $G = (V, E)$. Write

$$N(v) = \{w : vw \in E\}, \qquad N[v] = N(v) \cup \{v\}, \qquad N[A] = \bigcup_{a \in A} N[a].$$

For a nonempty set $A \subseteq V$, let

$$X(A) = \bigcap_{a \in A} N(a)$$

be its common *open* neighborhood. A set C dominates a request set W if $W \subseteq N[C]$. In particular, $X(\{v\}) = N(v)$. For $W \subseteq V$, put

$$e_G(W) = |E(G[W])|, \qquad \mathrm{def}_G(W) = 3|W| - e_G(W).$$

We omit the subscript G when the ambient graph is clear. The *coarse deficit* $\mathrm{def}(W)$, with $3|W|$ rather than $3|W| - 6$, is convenient for small and disconnected induced subgraphs. For a function $y : V \to \mathbb{R}$, write $y(W) = \sum_{w \in W} y_w$. All sums over empty sets are zero.

2.1 Planar drawings, edge bounds, and contractions

A graph is planar if it has a drawing on the sphere in which vertices are distinct points and edges are arcs meeting only at shared endpoints. The following standard facts about planar

drawings are included with proofs; see also Diestel [21, Chapter 4] for general background. A planar drawing is used only in the analysis, not supplied to the algorithm.

For disjoint A, B, let $e_G(A, B)$ count edges with one endpoint in each set. The graph consisting of these edges is bipartite, regardless of any other edges of G.

Fix a drawing of a connected graph with at least one edge. A *dart* is an oriented edge (u, v), and its reverse is (v, u). The drawing gives a cyclic order of the darts leaving each vertex. Let σ take each dart to its successor in this order, and put $\alpha(u, v) = (v, u)$. The cycles of

$$\varphi = \sigma \circ \alpha$$

are the facial boundary walks, traversed in one consistent direction. Every dart belongs to exactly one such cycle, so the sum of boundary lengths is $2e$. These walks may repeat vertices and may traverse both orientations of a bridge; they need not be simple cycles.

For a connected drawing with n vertices, e edges, and f faces,

$$n - e + f = 2. \tag{1}$$

For completeness, an embedded tree has one face: successively deleting leaves does not change the number of complementary regions and reduces to a single vertex. Since a tree has $n - 1$ edges, the formula holds. Otherwise, delete an edge on a cycle. The graph stays connected, and the two sides of the edge belong to distinct faces, since a simple cycle containing it separates the sphere. Deletion merges precisely those two faces, reducing both e and f by one. Repeating until a tree remains proves (1).

Lemma 2.1 (Planar edge bounds)**.** *A connected simple planar graph with $n \geq 3$ vertices has at most $3n - 6$ edges. If it is bipartite, it has at most $2n - 4$ edges. Consequently every finite simple planar graph, with no connectivity or minimum-order assumption, satisfies*

$$|E(G)| \leq 3|V(G)|, \qquad |E(G)| \leq 2|V(G)| \quad \text{if } G \text{ is bipartite.}$$

In particular, $\operatorname{def}(W) \geq 0$ *for every vertex set W, and $e_G(A, B) \leq 2(|A| + |B|)$ for disjoint A, B.*

Proof. A facial boundary cannot have length one, because that would give a loop. A boundary of length two has darts $(u, v), (v, u)$: successive darts form a walk, and simplicity permits only this return in two steps. The equalities

$$\varphi(u, v) = (v, u), \qquad \varphi(v, u) = (u, v)$$

say that the cyclic successor fixes (v, u) at v and (u, v) at u. A cyclic permutation fixes a member only if its fiber has size one. Hence both endpoints have degree one, so their connected component is K_2. Every facial boundary in a connected graph of order at least three therefore has length at least three. Thus $3f \leq 2e$, which together with (1) gives $e \leq 3n - 6$.

In a bipartite graph every closed walk has even length, since each step changes sides. For a connected component of order at least three, the preceding argument excludes length two, so every face has length at least four. The inequality $4f \leq 2e$ and (1) give $e \leq 2n - 4$.

Components of order one and two have respectively zero and at most one edge. They satisfy both relaxed bounds. Apply the connected bounds to all other components and sum; the empty graph also satisfies the claims. Induced subgraphs, and the bipartite subgraph between A and B, inherit planarity by deleting vertices and edges. This proves the final claims. □

Lemma 2.2 (Outside domination of a common neighborhood)**.** *For three distinct vertices a, b, d of a simple planar graph,*

$$|N(a) \cap N(b) \cap N[d]| \leq 3.$$

Proof. The graph $K_{3,3}$ is not planar: it is connected and bipartite, with six vertices and nine edges, whereas Lemma 2.1 allows at most eight. If the displayed intersection had at least four members, choose three distinct members w_1, w_2, w_3 different from d. Each is adjacent to a, b, d. None equals a or b, by absence of loops, and none equals d by choice. The two triples therefore form a $K_{3,3}$ subgraph. Deleting all other vertices and edges would preserve its planar drawing, a contradiction. □

Lemma 2.3 (Simultaneous contraction of domination stars)**.** *Let D dominate a planar graph G, and fix a map $f : V(G) \to D$ with*

$$f(v) \in N[v] \quad (v \in V(G)), \qquad f(d) = d \quad (d \in D).$$

For any $L \subseteq V(G) \setminus D$, put

$$q(v) = \begin{cases} f(v), & v \in L, \\ v, & v \notin L. \end{cases}$$

The simple graph F on $V(G) \setminus L$ with edge set

$$\{\{q(u), q(v)\} : uv \in E(G),\ q(u) \neq q(v)\}$$

is planar. All vertices outside L remain distinct. If $W \cap L$ is empty, every edge of $G[W]$ remains in $F[W]$.

Proof. Such an assignment exists by fixing D pointwise and choosing a dominating neighbor in D for every other vertex. If $v \in L$, then $v \neq f(v)$ and the edge $vf(v)$ exists. These edges form vertex-disjoint stars centered in D: every leaf has one assigned center, and no center is a leaf.

Contract all these star edges, retaining each center's name. Edge contraction preserves a planar drawing. Indeed, choose a narrow disk around an edge and its endpoints, containing no other vertex. The incident arcs can be redrawn inside this disk to meet one vertex in their existing boundary order, without crossings. Carry this out one edge at a time. Delete resulting loops, and keep one copy of each group of parallel edges. Neither deletion introduces a crossing. No two centers, and no two vertices outside L, are identified. The resulting vertex map is exactly q, and its simple edge set is the one stated. For $u, v \in W$ the map fixes both distinct endpoints, so each edge uv survives. □

Lemma 2.4 (Realizing assigned incidences)**.** *Keep D, f as above. Let $A \subseteq V(G) \setminus D$, $B \subseteq D$, and $P \subseteq A \times B$ be a set of distinct ordered pairs. Suppose each $(a, d) \in P$ has a witness $w_{a,d}$ with*

$$aw_{a,d} \in E(G), \qquad f(w_{a,d}) = d, \qquad w_{a,d} \notin A.$$

Then

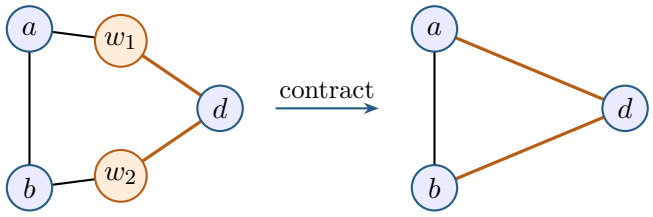
$$|P| \leq 2(|A| + |B|). \tag{2}$$

If also $A \cup B \subseteq W$, all witnesses lie outside W, and $ad \notin E(G)$ for every $(a, d) \in P$, then

$$|P| \leq \mathrm{def}(W). \tag{3}$$

Proof. Let L consist of the chosen witnesses outside D, and apply Lemma 2.3. No vertex of $A \cup B$ is contracted. Every witness maps to its assigned vertex d. This includes a witness already in D: it is then equal to $f(w) = d$ and is left fixed. The edge $aw_{a,d}$ consequently becomes ad, with distinct endpoints because $A \cap D$ is empty.

Different ordered pairs in P give different unordered edges: disjointness of A, B determines the orientation from A to B. Restricting to those edges gives a simple planar bipartite graph on $A \cup B$. Its edge bound proves (2).



Before: $f(w_1) = f(w_2) = d$ After: old edge ab is retained

Figure 2: Contraction of a domination star. Blue vertices are retained, and the orange witness vertices are contracted to their assigned dominator d. If the blue vertices lie in a protected set W and the witnesses lie outside it, every original edge of $G[W]$ survives. Distinct new edges inside W can then be charged to $\mathrm{def}(W) = 3|W| - e_G(W)$. The picture illustrates the operation, without assuming that this small example itself has a hard neighborhood.

Under the additional assumptions, the contractions fix W, so $F[W]$ contains every original edge of $G[W]$. The $|P|$ edges obtained above are all distinct and were absent from $G[W]$. Hence

$$e_G(W) + |P| \le |E(F[W])| \le 3|W|,$$

proving (3). □

Lemma 2.5 (Pair counts from endpoint degrees)**.** *Let $\mathcal{P}$ be a simple graph of pairs of maximum degree at most T. For any vertex set Z, at most $T|Z|/2$ of its edges have both endpoints in Z. If all its endpoints lie in $A \subseteq V(G)$, put*

$$c(w) = |\{\{a, b\} \in E(\mathcal{P}) : a, b \in N_G(w)\}|.$$

For planar G,

$$\sum_{w \in A} c(w) \le 3T|A|, \tag{4}$$

$$\sum_{w \in B} c(w) \le T(|A| + |B|) \quad \text{if } A \cap B = \varnothing. \tag{5}$$

The edges of $\mathcal{P}$ need not be edges of G.

Proof. The sum of degrees in $\mathcal{P}[Z]$ counts every internal pair twice and is at most $T|Z|$. Apply this to $Z = N_G(w) \cap A$. For $w \in A$, summing gives

$$\sum_{w \in A} c(w) \le \frac{T}{2} \sum_{w \in A} |N_G(w) \cap A| = T e_G(A) \le 3T|A|.$$

For disjoint A, B, it gives

$$\sum_{w \in B} c(w) \le \frac{T}{2} e_G(A, B) \le T(|A| + |B|).$$

Both last inequalities use Lemma 2.1. □

3 Exact maximal packing with bounded row size

Maximal fractional packings are a standard tool for local covering algorithms; see, for example, Åstrand and Suomela [16]. We need an exact packing result whose round bound depends on

constraint size and on a fixed finite set of capacities, but not on the number of constraints containing a variable. We prove that result here.

Let I be a finite set of variables and $\mathcal{A}$ a finite set of row indices. A row $r \in \mathcal{A}$ has a support $E_r \subseteq I$ and a capacity $c_r > 0$. Different rows may have the same support and capacity. There are no repeated occurrences of a variable within a row. Given a common cap $M > 0$, a vector $x \in \mathbb{R}^I$ is *feasible* if

$$0 \le x_i \le M \quad (i \in I), \qquad \sum_{i \in E_r} x_i \le c_r \quad (r \in \mathcal{A}). \tag{6}$$

It is *coordinatewise maximal* if there is no other feasible vector x' with $x'_i \ge x_i$ for every i and a strict inequality for at least one i. In the incidence network there is one node for each variable and one node for each row, with an edge ir exactly when $i \in E_r$. Each row node knows its capacity. Empty rows are harmless and may be discarded. We allow isolated variable nodes.

Lemma 3.1 (Exact bounded-arity packing)**.** *Fix an integer $K \ge 0$, a finite set $C \subseteq \mathbb{Q}_{>0}$, and $M \in \mathbb{Q}_{>0}$. There is an effectively constructible finite set*

$$\mathcal{P}(K, C, M) \subseteq [0, M] \cap \mathbb{Q}, \qquad 0, M \in \mathcal{P}(K, C, M),$$

with the following property. On every finite incidence network satisfying $|E_r| \le K$ and $c_r \in C$, a deterministic algorithm computes a coordinatewise maximal feasible vector using at most

$$2(|\mathcal{P}(K, C, M)| - 1)$$

communication rounds. Every output coordinate belongs to $\mathcal{P}(K, C, M)$. The algorithm needs neither the size of the instance nor a bound on the number of rows containing a variable. It uses exact rational arithmetic and does not require identifiers or symmetry breaking on the incidence network.

3.1 A certificate and a finite progressive-filling process

We call a feasible vector x a *water-filling certificate* if, for each $i \in I$, either $x_i = M$, or there is a row r such that

$$i \in E_r, \qquad \sum_{j \in E_r} x_j = c_r, \qquad x_j \le x_i \quad \text{for every } j \in E_r. \tag{7}$$

Thus a noncapped coordinate has a tight row whose other coordinates are no larger. This last condition is the additional information used in the palette argument.

Lemma 3.2. *Every finite system with positive rational capacities and positive rational common cap has a rational water-filling certificate. Every water-filling certificate is coordinatewise maximal.*

Proof. Here is a finite, global process used to prove existence. Its global minimum operations will not be part of the distributed algorithm. Initially all coordinates are active, their values are zero, and the current level is $a = 0$. Active coordinates always have the common value a; every already frozen coordinate has its final value, at most a.

Suppose the active set A is nonempty. For each row put

$$m_r = |E_r \cap A|, \qquad F_r = \sum_{i \in E_r \setminus A} x_i.$$

For rows with $m_r > 0$ define $b_r = (c_r - F_r)/m_r$, and let

$$b = \min(\{M\} \cup \{b_r : r \in \mathcal{A},\ m_r > 0\}). \tag{8}$$

The set in this minimum is finite and nonempty, and all its entries are rational. Current feasibility gives $b_r \geq a$ for every such row, and hence $a \leq b \leq M$. Raise every active coordinate to b and leave frozen coordinates unchanged. A row with $m_r > 0$ now has load $F_r + m_r b \leq c_r$; a row with $m_r = 0$ has not changed. Thus the new vector is feasible.

Freeze every active coordinate lying in any row that is now tight; also freeze all active coordinates if $b = M$. All simultaneous tight rows are processed in this step. At least one active coordinate freezes: either M attains the minimum and all of them freeze, or some row with $m_r > 0$ attains the minimum and all of its active coordinates freeze. For a coordinate frozen by a tight row, all coordinates of that row are now frozen, and their values are at most b. Consequently that row stays tight forever and supplies (7) for the coordinate just frozen. Previously supplied certificates are preserved. The level is now b, and the process repeats on the remaining active set.

This proves termination after at most $|I|$ steps. More explicitly, when any active coordinates remain, no tight row contains an active coordinate, since the preceding step froze all such coordinates. Also the current level is below M. Hence the next level in (8) is strictly larger than the current level. The initial positivity assumptions give the same conclusion for the first step. Every operation uses rational arithmetic, so the final vector is rational. If I is empty there is no step and the conclusion is vacuous. An isolated variable is eventually frozen at M. The final vector has the asserted certificate.

Finally, let x be any vector satisfying the certificate, and suppose $x' \geq x$ is feasible. If $x'_i > x_i$, then $x_i \neq M$. A tight row witnessing (7) for i has strictly larger load at x' than at x, contradicting feasibility. Thus no coordinate can increase, as required. □

3.2 An explicit finite palette

We now define $\mathcal{P}(K, C, M)$ recursively. This is a finite computation on rational numbers; it does not examine the incidence network. For $K = 0$, or for $C = \varnothing$, set

$$\mathcal{P}(K, C, M) = \{0, M\}. \tag{9}$$

For $K \geq 1$ and $C \neq \varnothing$, form the primary set

$$B = \{0, M\} \cup \left\{ \frac{c}{h} : c \in C,\ 1 \leq h \leq K,\ 0 < \frac{c}{h} < M \right\}. \tag{10}$$

Write its distinct elements in increasing order as $0 = b_0 < b_1 < \cdots < b_s = M$. Initialize $F = \{0\}$. Process the consecutive intervals $(a, b) = (b_j, b_{j+1})$ from left to right. At the beginning of an interval define the finite set

$$C_{a,F} = \mathbb{Q}_{>0} \cap \left\{ c - \sum_{\nu=1}^{d} p_\nu - ma : \begin{array}{l} c \in C, \quad d, m \geq 1, \quad d + m \leq K, \\ p_1, \ldots, p_d \in F \end{array} \right\}. \tag{11}$$

The notation in this display means that only positive values of the displayed expression are retained. Repetition among the p_ν is allowed. Compute

$$P' = \mathcal{P}(K - 1, C_{a,F}, b - a)$$

and replace F by

$$F \ \longleftarrow\ F \cup \{a + t : t \in P'\} \cup \{b\}. \tag{12}$$

After the last interval, return F as $\mathcal{P}(K, C, M)$.

There are finitely many primary intervals and finitely many choices in (11). Every recursive call decreases K; its new common cap $b - a$ is positive and its capacity set is finite and positive. The procedure therefore terminates. All comparisons, sorting, and arithmetic are

exact operations on rational numbers, so the palette is effectively constructible. By induction, the recursive palette P' lies in $[0, b-a]$, and its translate lies in $[a, b]$. At the start of an interval, $F \subseteq [0, a]$; after the update, $F \subseteq [0, b]$. It follows in particular that $\mathcal{P}(K, C, M) \subseteq [0, M] \cap \mathbb{Q}$, and that the final palette contains both 0 and M. These facts also cover an empty residual capacity set. No quantity in this construction depends on $|I|$, $|\mathcal{A}|$, or variable frequency.

Lemma 3.3 (Palette coverage)**.** *If $|E_r| \le K$, $c_r \in C$, and x is a water-filling certificate with common cap M, then*

$$x_i \in \mathcal{P}(K, C, M) \quad \textit{for every } i \in I.$$

Proof. We prove the assertion by induction on K, for every choice of finite positive rational C and positive rational M, and every finite system. If $K = 0$, no row can contain a variable. If $C = \varnothing$, there are no rows at all. In either case the certificate forces $x_i = M$ for every i, proving the assertion for the base cases.

Assume $K \ge 1$, $C \ne \varnothing$, and the assertion for $K-1$. In the primary-interval sweep maintain the invariant

$$x_i \le a \quad \Longrightarrow \quad x_i \in F. \tag{13}$$

It holds initially at $a = 0$, since feasibility makes every such value zero. Consider one consecutive primary interval (a, b). By the definition of the primary set, for every $c \in C$ and every $1 \le h \le K$ we have

$$\frac{c}{h} \le a \quad \text{or} \quad \frac{c}{h} \ge b. \tag{14}$$

Indeed, a ratio strictly between a and b would be an omitted primary point. Ratios at the boundaries are allowed in this statement; ratios at least M satisfy the second alternative.

Let $I_a = \{i \in I : x_i > a\}$. For each row split its support into

$$O_r = \{i \in E_r : x_i \le a\}, \qquad L_r = \{i \in E_r : x_i > a\}.$$

Keep only rows for which O_r and L_r are both nonempty. For every kept row define a residual row on I_a with support L_r and capacity

$$c'_r = c_r - \sum_{i \in O_r} x_i - |L_r| a. \tag{15}$$

Its arity is at most $K - 1$, because at least one old coordinate was deleted. Its capacity is positive: feasibility and $L_r \ne \varnothing$ give

$$c'_r \;\ge\; \sum_{i \in L_r} (x_i - a) \;>\; 0.$$

By the sweep invariant every old value belongs to F. Taking $d = |O_r|$, $m = |L_r|$ in (11) therefore shows that $c'_r \in C_{a,F}$. In particular the residual capacities are rational, even if we initially allowed a real-valued certificate.

Give this residual system the common cap $b - a$, and set

$$z_i = \min\{x_i, b\} - a \quad (i \in I_a). \tag{16}$$

We claim that z is a water-filling certificate for the residual system. First, $0 < z_i \le b - a$, and each retained row satisfies

$$\begin{aligned} \sum_{i \in L_r} z_i &= \sum_{i \in L_r} \min\{x_i, b\} - |L_r| a \\ &\le c_r - \sum_{i \in O_r} x_i - |L_r| a = c'_r. \end{aligned}$$

Thus z is feasible.

If $x_i \geq b$, then $z_i = b - a$, so the common-cap alternative of the certificate holds. It remains to consider $a < x_i < b$. Since $b \leq M$, this coordinate is not capped in the original system. Choose a witness row r from (7), and put $h = |E_r|$. It contains i, so $1 \leq h \leq K$, and

$$\frac{c_r}{h} = \frac{1}{h} \sum_{j \in E_r} x_j \leq x_i < b.$$

Equation (14) gives $c_r/h \leq a$. If O_r were empty, all h coordinates of this row would be strictly greater than a, and its tightness would yield $c_r > ha \geq c_r$, a contradiction. Hence $O_r \neq \varnothing$; also $i \in L_r$, so this row was retained. Every original coordinate in the witness row is at most $x_i < b$. No one of them is altered by truncation at b, and therefore

$$\sum_{j \in L_r} z_j = \sum_{j \in L_r} x_j - |L_r|a = c'_r.$$

Furthermore, $z_j = x_j - a \leq x_i - a = z_i$ for every $j \in L_r$. This is precisely the required residual certificate.

The induction hypothesis now applies to the entire residual system, including the case of no residual rows or no residual variables. Thus $z_i \in \mathcal{P}(K-1, C_{a,F}, b-a)$ for each $i \in I_a$. If $a < x_i < b$, then $x_i = a + z_i$ and the update (12) lists x_i. Values $x_i \leq a$ remain in F, and the boundary value $x_i = b$ is added explicitly. We have proved the sweep invariant with b in place of a. After the final interval we have $b = M$, and feasibility gives $x_i \leq M$ for every coordinate. This completes the induction. □

Two features of the interval argument are useful to make explicit. A row all of whose coordinates have value at most a has no residual variable and cannot supply a new interior event. A row with no such coordinate cannot witness an event in (a, b): the average-capacity argument in the proof forces every witness row for such an event to contain an old coordinate. Thus only rows with both types are needed in the residual system. Also, the artificial cap $b - a$ only truncates coordinates whose original final values are at least b. It does not assert that those original coordinates freeze at b. A value equal to b is covered separately, and later intervals handle values above b. These observations apply equally at the first interval, at the last interval, and when some primary points coincide before duplicates are removed.

3.3 The fixed scan and its locality

Sort the complete palette as

$$\mathcal{P}(K, C, M) = \{0 = t_0 < t_1 < \cdots < t_L = M\}.$$

The incidence-network algorithm begins with value zero at every variable and with all variables unfrozen. For each $h = 1, \ldots, L$, it performs the following two communication phases.

1. Every unfrozen variable changes its value to t_h; a frozen variable keeps its current value. Every variable sends its value to each incident row.

2. Every row computes its exact load and reports to every incident variable whether the row is tight. A variable freezes if it receives a tight report or if $t_h = M$.

Variables already frozen remain frozen. A variable with no incident row receives no report and freezes at the common cap. The scan uses a fixed finite schedule independent of the instance; it does not wait for a globally detected next event.

Proof of Lemma 3.1. Let x^* be a certificate supplied by Lemma 3.2. Every coordinate of x^* is positive: if $x_i^* = 0$, then it is below M, and a witness row would have all its nonnegative coordinates at most zero, contradicting its positive capacity. Lemma 3.3 gives $x_i^* \in \mathcal{P}(K, C, M)$. We prove by induction on the scan levels that after level t_h the current value at i is

$$\min\{x_i^*, t_h\}, \qquad \text{and } i \text{ is frozen exactly when } x_i^* \le t_h. \tag{17}$$

At $t_0 = 0$ this follows from positivity and initialization. Suppose it holds after level $a = t_{h-1}$ and write $t = t_h$. There is no coordinate value of x^* strictly between a and t, because both are consecutive points of a palette containing all such values. A coordinate frozen at the previous level has its final value $x_i^* \le a$. An unfrozen coordinate has $x_i^* > a$, hence $x_i^* \ge t$, and is raised to t. Thus after the first phase the current vector is exactly $\min\{x^*, t\}$ coordinate by coordinate. This vector is feasible, since it is nonnegative and coordinatewise no larger than the feasible vector x^*.

If $x_i^* > t$, then $t < M$. In every row containing i, the current load is strictly smaller than the final load: this coordinate has been strictly decreased, and no other coordinate has increased. The final load is at most capacity, so no incident row is tight now. Thus i does not freeze prematurely. If $x_i^* = t$, then i freezes at the common cap when $t = M$. Otherwise, its witness row in (7) has every final coordinate at most t. The current values on that row already equal their final values, and the row is tight, so i freezes in this step. Earlier-frozen coordinates stay frozen. This proves the full invariant.

At the last level M, the output is exactly x^*. It is rational, feasible, and coordinatewise maximal by Lemma 3.2. The number of phases is $2L = 2(|\mathcal{P}| - 1)$. Each incidence edge carries one message in each direction per level. A variable may have arbitrarily many incident rows, but all its messages are sent simultaneously. The LOCAL model permits an arbitrarily large finite message on every edge and unbounded local computation, so this does not alter the round count. No identifiers, orientation, or ordering of incident edges enter the algorithm. This proves the lemma. □

3.4 Effective arithmetic

The construction above gives a finite algorithm, rather than only the existence of a suitable finite set of numbers. The recursion for the palette terminates by decreasing the nonnegative integer K, and its inner loops enumerate finite sets and tuples of rational numbers. It can therefore be executed as a preliminary computation depending only on the fixed parameters, or its finite output can be included in the program. Neither alternative requires information about the network.

To implement all scan operations using integers, take any positive integer D divisible by the denominators of every member of $\mathcal{P}(K, C, M) \cup C \cup \{M\}$. Such a D is obtained, for example, by multiplying the positive denominators in reduced fraction representations. Multiply every value, capacity, and scan level by D. Every variable then stores a nonnegative integer. Row loads are finite sums of integers, and tightness is an integer equality. This also shows that exact comparisons cause no issue of real-number computation. The size of the palette and of D may be enormous; only their finiteness and independence from the network are required.

Remark 3.4. The constant in Lemma 3.1 depends on row size and on a fixed finite set of rational capacities; it need not be efficient in these parameters. This is consistent with the $\Omega(\Delta)$ locality barrier for maximal fractional matching when the running time is independent of the network size [17]: its constraints are indexed by vertices, its variables by edges, and its row size can be the maximum degree Δ. The lemma makes no half-integrality claim. Fine rational denominators can be necessary in local maximal fractional matching [18].

4 The hard-neighborhood reduction and the algorithm

Fix the following constants for the rest of the paper:

$$\theta = \frac{1}{7}, \qquad T = 10^6, \qquad Q = 10^{12}. \tag{18}$$

In particular, $1 \le T \le Q$.

4.1 Hard sets and the retained hypergraph

A hypergraph here is simply a family of nonempty vertex sets, called hyperedges. Its rank is the maximum hyperedge size, and a vertex's hypergraph degree is the number of hyperedges containing it.

Definition 4.1. A nonempty set $A \subseteq V$ is *q-hard* if no set $C \subseteq V \setminus A$ with $|C| \le q$ dominates $X(A)$.

Let $\mathcal{H}_0$ consist of the Q-hard singletons and the Q-hard two-element sets. For each vertex v, put

$$P(v) = \{u \ne v : \{v, u\} \in \mathcal{H}_0\}.$$

Define the singleton centers and the retained hypergraph by

$$C_* = \{v : \{v\} \in \mathcal{H}_0 \text{ or } |P(v)| > T\}, \tag{19}$$

$$\mathcal{H} = \{\{v\} : v \in C_*\} \ \cup \ \{e \in \mathcal{H}_0 : |e| = 2, \ e \cap C_* = \varnothing\}. \tag{20}$$

Thus we add a singleton at every vertex incident with more than T original hard pairs, and remove every pair that contains a singleton center. The pairs sharing a center have disjoint one-vertex remainders, the rank-two instance of a sunflower [20]. Its elementary reduction properties are proved next.

Set

$$X = \bigcup_{e \in \mathcal{H}} X(e), \qquad R = V \setminus X. \tag{21}$$

The vertices of R are the *residual requests*.

Lemma 4.2. *The construction satisfies the following assertions.*

(i) Every retained hyperedge has one or two vertices, every hard set has a nonempty common neighborhood, and the endpoints of every retained pair are at distance at most two.

(ii) Every edge of $\mathcal{H}_0$ contains an edge of $\mathcal{H}$. Every retained singleton is T-hard, and every retained pair is Q-hard.

(iii) Every vertex belongs to at most T hyperedges of $\mathcal{H}$.

(iv) If $1 \le |A| \le 2$ and $X(A) \cap R \ne \varnothing$, then A is not Q-hard.

(v) For every $v \in V$,

$$|N(v) \cap R| \le B_0, \qquad B_0 := Q(3Q + 2).$$

Proof. If $X(A) = \varnothing$, the empty set dominates it, so A is not hard. This proves the nonemptiness assertion and the distance bound for pairs. A pair of $\mathcal{H}_0$ is either retained or contains a center of C_*; an original singleton is retained. Hence every original edge contains a retained edge.

An original hard singleton is T-hard because $T \le Q$. Suppose that v is a new center with $|P(v)| > T$, and that $C \subseteq V \setminus \{v\}$, $|C| \le T$, dominates $N(v)$. For every $u \in P(v)$, the set C dominates $X(\{v, u\}) \subseteq N(v)$. If $u \notin C$, this is a cover of that common neighborhood from

outside $\{v, u\}$, contradicting its Q-hardness. Thus $P(v) \subseteq C$, which is impossible. Retained pairs are original hard pairs.

A center belongs only to its own singleton in $\mathcal{H}$. A noncenter belongs only to original pairs, of which it has at most T. For (iv), if A were Q-hard, it would contain a retained hyperedge e, and $X(A) \subseteq X(e) \subseteq X$, contrary to $X(A) \cap R \neq \varnothing$.

For (v), write $m = |N(v) \cap R|$, and suppose $m > Q(3Q+2)$. By (iv) applied to $\{v\}$, some set $C \subseteq V \setminus \{v\}$, $|C| \leq Q$, dominates $N(v)$. For $a \in C$, at most one vertex of $N(v) \cap R$ is dominated by equality with a; all other vertices covered by a are in $N(v) \cap N(a) \cap R$. If these intersections all had at most $3Q$ vertices, their total cover would have size at most

$$|C|(3Q+1) \leq Q(3Q+1) < m.$$

Consequently some $a \in C$ satisfies $|N(v) \cap N(a) \cap R| > 3Q$. In particular $a \neq v$, and (iv) applied to $\{v, a\}$ gives an outside cover of $X(\{v, a\})$ of size at most Q. By the planar three-neighbor bound (Lemma 2.2), one vertex outside this pair dominates at most three of its common neighbors. Such a cover cannot dominate more than $3Q$ common neighbors, a contradiction. □

4.2 The mixed packing and the output

For each $w \in R$ introduce a variable y_w, and for each $e \in \mathcal{H}$ a variable z_e. Compute a coordinatewise maximal feasible vector for the following packing constraints:

$$\begin{aligned} \ell_v := \sum_{w \in R \cap N[v]} y_w + \sum_{\substack{e \in \mathcal{H} \\ v \in e}} z_e \leq 1 \qquad & (v \in V), \\ 0 \leq y_w \leq \theta \qquad & (w \in R), \\ 0 \leq z_e \leq 1 \qquad & (e \in \mathcal{H}). \end{aligned} \tag{22}$$

The upper bounds $z_e \leq 1$ are redundant, because every hyperedge is nonempty and occurs in the unit row of each endpoint. We state them explicitly to use a common coordinate cap $M = 1$. The bounds $y_w \leq \theta$ are singleton rows of capacity θ. By Lemma 4.2, every unit row has at most

$$K := B_0 + 1 + T = Q(3Q+2) + 1 + T$$

variables. All row capacities belong to $\{1, 1/7\}$. Thus Lemma 3.1 supplies an exact maximal packing, with rational coordinates, using a fixed finite list of levels. Extend y by setting $y_v = 0$ for $v \notin R$.

Define

$$S = \left\{ v : \ell_v \geq 1 - \frac{\theta - y_v}{2} \right\}, \qquad U = R \setminus N[S], \qquad O = S \cup U. \tag{23}$$

The threshold depends on y_v. Selecting only the exactly tight rows would lose a slack term that will be needed later.

Lemma 4.3. *The set O is dominating. Moreover, $S \cap U = \varnothing$, every $e \in \mathcal{H}$ meets S, and*

$$y_w = \theta \quad \text{for every } w \in U. \tag{24}$$

Proof. A tight unit row belongs to S, since $y_v \leq \theta$. Consider a hyperedge e. If $z_e = 1$, every endpoint has a tight unit row. If $z_e < 1$ and no endpoint row is tight, then all rows containing z_e have positive slack. There are finitely many such rows, so z_e could be increased by some positive amount, contrary to coordinatewise maximality. Thus $e \cap S \neq \varnothing$.

Now let $w \in U$. None of the unit rows containing y_w is tight: such a row is indexed by a vertex of $N[w]$, which would belong to S and dominate w. If $y_w < \theta$, its personal cap and all

its incident rows therefore have positive slack, again allowing a positive increase. This proves (24).

The definition of U gives $S \cap U = \varnothing$ and ensures that O dominates R. For every $w \in X$, there is an $e \in \mathcal{H}$ with $w \in X(e)$; any vertex in $e \cap S$ is adjacent to w. Hence O also dominates X, and $R \cup X = V$. □

4.3 An effective constant-round implementation

The preprocessing takes at most six rounds. We give an explicit schedule, including the initial exchange of identifiers. In the first round, every vertex sends its identifier to all its neighbors. In the second, it sends its complete list of neighbor identifiers. After these rounds a vertex v knows $N(v)$ and $N(w)$ for every $w \in N(v)$. In particular it knows every vertex at distance at most two from v, and every edge incident with a neighbor of v.

This information determines all original hard sets containing v. For a singleton the request set is $N(v)$. For a candidate pair $\{v, u\}$, it is

$$X(\{v, u\}) = \{w \in N(v) : u \in N(w)\},$$

which v can compute from the received lists. Only vertices $u \neq v$ at distance at most two need be considered: a pair with no common neighbor is not hard, and every pair with a common neighbor has this distance bound. For either kind of tested set A containing v, put $W = X(A) \subseteq N(v)$. Any vertex that dominates at least one request of W lies in

$$N[W] = \bigcup_{w \in W} N[w] \subseteq B_2(v),$$

where $B_2(v)$ is the closed distance-two ball. In a proposed cover, vertices outside $N[W]$ cover no request and can be deleted. Thus v can decide Q-hardness by enumerating all subsets of $N[W] \setminus A$ of size at most Q and testing whether they cover W. The received neighbor lists give exactly the incidences needed for every such test. This also handles $W = \varnothing$, for which the empty cover succeeds. Consequently, by the end of round two, v knows its complete hard-partner set $P(v)$ and whether it belongs to the singleton-center set C_*.

In rounds three and four, vertices forward their center bits, tagged with their identifiers, to distance two. Every v then knows the center bits of all $u \in P(v)$, so it can determine its complete list of incident retained hyperedges: its singleton if $v \in C_*$, and each original hard pair whose two endpoints lie outside C_*. All endpoints obtain consistent descriptions because both hardness and the two-center test are symmetric.

In round five, every vertex sends its list of incident retained hyperedges to all its graph neighbors. A receiving vertex w is in X exactly when one of the received hyperedges e satisfies $e \subseteq N(w)$. The forward implication follows because every endpoint of such an e is a neighbor of w and sends its incident edge list; the reverse implication is the definition of $X(e)$. Thus w determines whether it belongs to $R = V \setminus X$. In round six, vertices exchange their residual-membership bits. Every vertex v now knows $R \cap N[v]$, all retained hyperedges incident with v, and hence its complete mixed packing row.

The length-one or length-two routes used for pair variables are already available from the information obtained in round two. For a pair at distance two, both endpoints know its common neighbors and can choose the one with smallest identifier. All messages and enumerations in this schedule are finite; their size need not be bounded. The schedule applies to every finite graph, without knowing its order or maximum degree and without testing planarity.

Simulating the packing incidence network. We apply the packing lemma to the residual vertex set R and the hypergraph $\mathcal{H}$. The properties needed for this application are

$$1 \le |e| \le 2, \qquad \operatorname{diam}_G(e) \le 2 \quad (e \in \mathcal{H}),$$
$$|\{e \in \mathcal{H} : v \in e\}| \le T, \qquad |N(v) \cap R| \le B_0 = Q(3Q+2) \quad (v \in V(G)).$$

The variables are y_w for $w \in R$ and z_e for $e \in \mathcal{H}$. A vertex row has the form

$$\ell_v = \sum_{w \in R \cap N[v]} y_w + \sum_{e \in \mathcal{H} : v \in e} z_e \le 1. \tag{25}$$

In addition there is one personal row $y_w \le 1/7$ for each $w \in R$. All variables are nonnegative. Add the common cap $M = 1$. This cap is redundant for y_w because of its personal row, and for z_e because $e \neq \varnothing$: a unit-capacity row at any endpoint of e contains z_e, and all other terms in that row are nonnegative. The vertex rows have at most $B_0 + 1 + T$ variables and the personal rows have one. Thus Lemma 3.1 applies with the fixed parameters

$$K = B_0 + 1 + T, \qquad C = \{1, 1/7\}, \qquad M = 1. \tag{26}$$

Let $L = |\mathcal{P}(K, C, 1)| - 1$ for these parameters.

Host the variable y_w, its personal row, and the vertex row at w at the graph vertex w. Host z_e at the endpoint of e with smallest identifier. Vertex row v is hosted at v, whether or not $v \in R$. Every incidence of a y variable with a vertex row has host distance at most one, and the personal row is at the same host. Every incidence of z_e with a vertex row has host distance at most two, by the diameter property of e.

The local preprocessing supplies these incidences and their endpoint identifiers. For a pair at distance two, choose a length-two route deterministically, for example through its common neighbor of smallest identifier; use its edge if the two endpoints are adjacent. Constant radius neighborhood information suffices to choose and record these routes. A message of the virtual incidence network is tagged with its variable and row names and sent along the chosen route. Every virtual communication phase takes at most two graph rounds. If many virtual messages use the same physical edge, concatenate them into one finite message. This is permitted in LOCAL and creates no additional rounds. Messages between nodes hosted at the same vertex require no communication. Consequently the two phases of each palette level take at most four graph rounds, for a total of at most $4L$ rounds for the packing scan.

At the end of the scan, each vertex row host knows its exact load ℓ_v, and each vertex knows its own y_v, with $y_v = 0$ outside R. It can decide locally whether

$$v \in S \quad \Longleftrightarrow \quad \ell_v \ge 1 - \tfrac{1}{2}(1/7 - y_v).$$

After one exchange of these membership bits, each vertex can decide whether it belongs to $U = R \setminus N[S]$ and hence to $O = S \cup U$. Using at most ten rounds for the constant-radius preprocessing, and allowing another ten rounds for route initialization and final membership exchanges, the safe total bound is $4L + 20$. Its numerical value is a single constant, since Q and T are fixed. No vertex is given n, the maximum degree, or a planar embedding.

All preprocessing tests enumerate finite vertex sets within fixed radius neighborhoods, with the thresholds Q and T fixed. Although these neighborhoods may have unbounded size, they are finite, so the tests are computable using the unbounded local computation allowed by the model. The rational packing arithmetic can be put on the common integer grid described above. If $Y_v = Dy_v$ and $L_v = D\ell_v$, the selection test is the integer inequality

$$13D + 7Y_v \le 14L_v.$$

Thus the construction is an actual deterministic LOCAL program with finite descriptions of its constants. The same fixed program is defined on every finite input graph; its correctness and approximation guarantees use the planar preprocessing bounds. It never needs to test whether its input is planar.

5 The slack available in rounding

The analysis compares the output with an arbitrary dominating set D. The algorithm never computes or receives this set. We first record the slack furnished by rounding, and then show that almost all vertices of D are suitable endpoints for planar contractions.

5.1 Finite level sums

All weights are rational. Choose a positive integer h, divisible by 14, such that hy_v is an integer for every vertex. This common denominator is used only in the analysis. For a function F on the positive grid points and $a \in \{1/14, 1/7\}$, write

$$\langle F(t)\rangle_a := \frac{1}{h}\sum_{j=1}^{ah} F(j/h). \tag{27}$$

This is a finite sum, so it is linear and preserves inequalities. For a constant c, $\langle c\rangle_a = ac$. For $p \in [0,\theta]$ with hp integral,

$$\left\langle \mathbf{1}_{\{t\le p\}}\right\rangle_\theta = p. \tag{28}$$

Indeed, exactly hp of the grid points j/h, $1 \le j \le h\theta$, are at most p. Thus this notation is a finite version of the usual layer-cake argument; no limiting or integration theorem is needed. If $F(t) \ge 0$, then $\langle F(t)\rangle_{1/14} \le \langle F(t)\rangle_\theta$.

Put

$$\begin{gathered} Y = \sum_{v\in V} y_v, \qquad Z = \sum_{e\in\mathcal{H}} z_e, \qquad \delta = \sum_{s\in S}(\theta - y_s), \\ B = N[S]\setminus S, \qquad Y_B = y(B), \\ W_t = S \cup \{v \in B : y_v \ge t\}, \qquad A_t = \{s \in S : y_s \ge t\}, \\ J = \langle \mathrm{def}(W_t) + \mathrm{def}(A_t)\rangle_\theta\,, \qquad J_0 = \langle \mathrm{def}(W_t)\rangle_\theta\,. \end{gathered} \tag{29}$$

All these quantities are nonnegative, and $J_0 \le J$. Every vertex with positive weight lies in exactly one of S, B, U: weights vanish off R, and $R \setminus N[S] = U$. By Lemma 4.3, $y(U) = \theta|U|$ and $S \cap U = \varnothing$. Also $y(S) = \theta|S| - \delta$. Consequently

$$7Y - |O| = 7Y_B - 7\delta. \tag{30}$$

Lemma 5.1 (Rounding with retained slack)**.** *The algorithm's output satisfies*

$$7Y - |O| \ge -\frac{13}{3}Z + \frac{7}{12}\delta + \frac{1}{2}Y_B + \frac{13}{6}J. \tag{31}$$

Proof. Consider the weighted neighbor incidence sum

$$M_S = \sum_{s\in S}\sum_{v\in N(s)} y_v.$$

The selection condition in (23) and the row definition give

$$\sum_{v\in N(s)} y_v \ge 1 - \theta + \frac{\theta - y_s}{2} - \sum_{\substack{e\in\mathcal{H}\\ s\in e}} z_e.$$

Each hyperedge has at most two endpoints, so summing yields

$$M_S \ge \frac{6}{7}|S| + \frac{1}{2}\delta - 2Z. \tag{32}$$

For the upper bound, count the contribution of each graph edge to $\langle e_G(W_t) + e_G(A_t)\rangle_\theta$. An edge uv with both endpoints in S contributes $\theta + \min\{y_u, y_v\}$, which is at least $y_u + y_v$, since $\max\{y_u, y_v\} \le \theta$. This pays its two contributions to M_S. An edge from S to a vertex $v \in B$ contributes exactly y_v, paying its sole contribution to M_S. All other edge contributions are nonnegative, and these two kinds account for every term in M_S. Using (28) and the definition of the deficit, we get

$$\begin{aligned} M_S &\le \langle e_G(W_t) + e_G(A_t)\rangle_\theta \\ &= 3\,\langle |W_t| + |A_t|\rangle_\theta - J \\ &= 3(\theta|S| + Y_B + y(S)) - J \\ &= \frac{6}{7}|S| + 3Y_B - 3\delta - J. \end{aligned} \tag{33}$$

Comparing (32) and (33) cancels the $|S|$ term and proves

$$3Y_B \ge \frac{7}{2}\delta + J - 2Z.$$

Substitute this in (30) to obtain

$$7Y - |O| \ge -\frac{14}{3}Z + \frac{7}{6}\delta + \frac{7}{3}J.$$

Take 13/14 of this inequality and add 1/14 of the exact identity (30). The left side is unchanged. The coefficients on the right become respectively $-13/3$, $7/12$, $1/2$, $13/6$, proving the assertion. □

5.2 Comparison with an integral dominating set

Fix any dominating set D. Separate the hyperedge mass into

$$Z_h = \sum_{\substack{e\in\mathcal{H}\\ e\cap D\neq\varnothing}} z_e, \qquad Z_m = \sum_{\substack{e\in\mathcal{H}\\ e\cap D=\varnothing}} z_e, \qquad Z = Z_h + Z_m.$$

Here the subscripts mean *hit* and *missed*. Define the unused capacity and the two overcount terms by

$$\begin{aligned} L_D &= \sum_{d\in D}(1-\ell_d), \\ M_y &= \sum_{w\in R} y_w\big(|N[w]\cap D| - 1\big), \\ M_z &= \sum_{\substack{e\in\mathcal{H}\\ e\cap D\neq\varnothing}} z_e\big(|e\cap D| - 1\big), \\ K_D &= L_D + M_y + M_z. \end{aligned} \tag{34}$$

All terms are nonnegative: rows are feasible, D dominates every vertex, and a hit hyperedge has at least one endpoint in D. Interchanging finite sums in the rows of D gives

$$\sum_{d\in D}\ell_d = \sum_{w\in R} y_w|N[w]\cap D| + \sum_{e\in\mathcal{H}} z_e|e\cap D| = Y + M_y + Z_h + M_z.$$

It follows that

$$|D| = Y + Z_h + K_D. \tag{35}$$

This is an exact accounting identity, not an assertion that the packing is optimal.

Define the common budget

$$\mathcal{R} = K_D + Z_h + \delta + Y_B + J_0 \geq 0. \tag{36}$$

Combining Lemma 5.1 with (35) gives

$$\begin{aligned} 7|D| - |O| &\geq 7K_D + \frac{8}{3}Z_h + \frac{7}{12}\delta + \frac{1}{2}Y_B + \frac{13}{6}J - \frac{13}{3}Z_m \\ &\geq \frac{1}{2}\mathcal{R} - \frac{13}{3}Z_m. \end{aligned} \tag{37}$$

For the second inequality, use $J \geq J_0$ and the fact that each of $7, 8/3, 7/12, 1/2, 13/6$ is at least $1/2$. We will prove that $Z_m \leq \mathcal{R}/100$.

5.3 Exceptional vertices of the comparison set

Let

$$D_b = \{d \in D : y_d < 1/14\} \cup (D \cap U), \qquad D_g = D \setminus D_b, \qquad S_0 = \{s \in S : y_s = 0\}. \tag{38}$$

The following bounds make these exceptional sets affordable.

Lemma 5.2. *For every dominating set D,*

$$|D_b| \leq 35\mathcal{R}, \qquad |S_0| \leq 7\mathcal{R}. \tag{39}$$

Moreover, $D_g \subseteq W_t$ and $y_d > 0$ for every $d \in D_g$ and every $0 < t \leq 1/14$.

Proof. If $s \in S$ has $y_s < 1/14$, its contribution $\theta - y_s$ to δ is greater than $1/14$. If $d \in D \setminus S$ has $y_d < 1/14$, the failure of the selection condition gives

$$1 - \ell_d > \frac{\theta - y_d}{2} > \frac{1}{28}.$$

Thus

$$|\{d \in D : y_d < 1/14\}| \leq 14\delta + 28L_D. \tag{40}$$

For $d \in D \cap U$, there is no member of S in $N[d]$. There are at most six members of U in $N[d]$: seven such vertices would, by $y_u = 1/7$, force $\ell_d \geq 1$, making $d \in S$, a contradiction. Every positive-weight vertex in $N[d]$ therefore belongs to U or B, and

$$\ell_d \leq \frac{6}{7} + \sum_{w \in B \cap N[d]} y_w + \sum_{\substack{e \in \mathcal{H} \\ d \in e}} z_e.$$

Rearrange and sum over $d \in D \cap U$. The unused row capacities sum to at most L_D. The boundary terms sum to at most $Y_B + M_y$: for each $w \in B$, its multiplicity is at most $|N[w] \cap D|$, and a positive weight implies $w \in R$. The hyperedge terms sum to at most $Z_h + M_z$, by the definition of M_z. Consequently

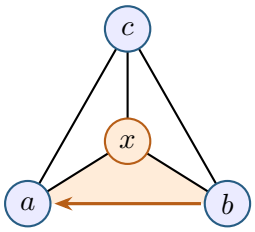
$$|D \cap U| \leq 7(L_D + Y_B + M_y + Z_h + M_z) = 7(K_D + Y_B + Z_h). \tag{41}$$

Combining (40) and (41), and using $L_D \leq K_D$, gives

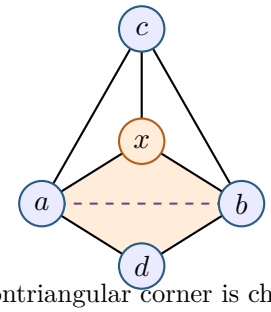
$$|D_b| \leq 14\delta + 35K_D + 7Y_B + 7Z_h \leq 35\mathcal{R}.$$

Each member of S_0 contributes $1/7$ to δ, so $|S_0| \leq 7\delta \leq 7\mathcal{R}$.

Finally, a member $d \in D_g$ has $y_d \geq 1/14 > 0$, hence lies in R. Since it is not in $U = R \setminus N[S]$, it lies in $N[S]$. It therefore belongs either to S or to $\{v \in B : y_v \geq t\}$ for every $0 < t \leq 1/14$. This proves $D_g \subseteq W_t$. □



A triangular corner identifies x from the oriented side (b, a)



A nontriangular corner is charged to the planar deficit

Figure 3: The two types of degree-three witnesses in the face count. Blue vertices belong to W; the orange vertex x lies outside W. On the left, the face with boundary a, x, b, a assigns the pair incidence to the dart (b, a), whose next facial dart ends at x. On the right, the shaded face has boundary x, a, d, b, x. The dashed segment ab denotes a pair in $\mathcal{P}$, not an edge of the planar graph. This distinction is allowed in the lemma.

6 A planar count for many pairs

Lemma 6.1 (Nontriangular darts, including small components)**.** *Let J be a finite simple planar graph. Draw its connected components separately on spheres, and use their facial permutations $\varphi = \sigma \circ \alpha$ as above. Let b be the number of darts on facial boundaries whose length is not three. Then, with $\Delta_J = 3|V(J)| - |E(J)|$,*

$$b \le 4\Delta_J. \tag{42}$$

Proof. For a connected component with $n \ge 3$ vertices, all facial boundaries have length at least three. A nontriangular one has length $\ell \ge 4$, so $\ell \le 4(\ell - 3)$. Summing, and including zero contributions from triangles, gives

$$b_{\text{comp}} \le 4 \sum_{\text{faces}} (\ell - 3) = 4(2e - 3f) = 4(3n - e - 6) \le 4(3n - e).$$

The equality uses Euler's formula. A component on two vertices is K_2. Its two darts form one boundary of length two, so $b_{\text{comp}} = 2 \le 4(6 - 1)$. An isolated vertex has no darts and satisfies $b_{\text{comp}} = 0 \le 4 \cdot 3$. Sum over components; both b and $3n - e$ are additive. The empty graph has both sides zero. □

Lemma 6.2 (Triangular corners of degree-three vertices)**.** *Fix planar rotations of J. Let $\mathcal{P}$ be any graph of pairs on $V(J)$ with m edges. Let C_3 consist of vertices of degree three in J for which every incident corner belongs to a triangular face. Then*

$$|\{(x, \{a, b\}) : x \in C_3,\ \{a, b\} \in E(\mathcal{P}),\ a, b \in N_J(x)\}| \le 2m.$$

Proof. Represent a corner at x by an incoming dart $q = (a, x)$. If (x, b) follows (x, a) in the cyclic order at x, then $\varphi q = (x, b)$. The corner is triangular precisely when q lies in a face cycle of length three.

Since x has three neighbors, their cyclic successor is a three-cycle. For any distinct neighbors a, b, exactly one follows the other. Orient their names so that a is followed by b, and take $q = (a, x)$. The triangular corner gives

$$q = (a, x), \qquad \varphi q = (x, b), \qquad p := \varphi^2 q = (b, a), \qquad \varphi p = q.$$

In particular, the opposite side ab is an edge of J. Map the incidence $(x, \{a, b\})$ to this oriented side $p = (b, a)$. The dart p recovers its underlying unordered pair and recovers x as the final

endpoint of φp. Thus the map is injective. Each of the m edges of $\mathcal{P}$ has only two orientations. A pair that is not an edge of J cannot occur as an image. The asserted bound follows. □

Lemma 6.3 (Ambient pair incidence)**.** *Let W, C be disjoint vertex sets in a finite simple planar graph G. Suppose every $x \in C$ has at least three neighbors in W. Let $\mathcal{P}$ be a graph of pairs on W with m edges and maximum degree at most T; its edges need not belong to G. Then*

$$\sum_{\{a,b\}\in E(\mathcal{P})} |N_G(a) \cap N_G(b) \cap C| \le 2m + 6T \operatorname{def}(W). \tag{43}$$

In particular, the inequality holds with $12T$ in place of $6T$. No connectivity or minimum-order assumption is required.

Proof. Let J have vertex set $W \cup C$, all edges of $G[W]$, and all edges between W and C. This deletes any edges inside C, so J is planar, C is independent, and $d_x := \deg_J(x) = |N_G(x) \cap W| \ge 3$ for $x \in C$. The incidence count is unchanged; by interchanging finite sums it is

$$I = \sum_{x\in C} c_J(x), \qquad c_J(x) = |\{\{a, b\} \in E(\mathcal{P}) : a, b \in N_J(x)\}|.$$

Set

$$E_C = \sum_{x\in C} (d_x - 3) \ge 0, \qquad \Delta_J = 3|V(J)| - |E(J)| \ge 0.$$

Every edge of J is either inside W or has exactly one endpoint in C. Hence

$$|E(J)| = e_G(W) + \sum_{x\in C} d_x, \qquad \operatorname{def}(W) = \Delta_J + E_C. \tag{*}$$

Fix componentwise planar rotations, and let b be the number of nontriangular darts. Partition C into: degree-three vertices whose corners are all triangular; degree-three vertices having a nontriangular corner; and vertices of degree at least four. The first class contributes at most $2m$ by Lemma 6.2.

For each vertex x of the second class, choose an incoming dart $q_x = (a_x, x)$ in a nontriangular corner. These darts are distinct, because their final endpoints are distinct. Thus there are at most b vertices in this class. Each contributes at most $3T/2$, by Lemma 2.5 applied to its three neighbors.

For $d_x \ge 4$, the same bound gives

$$c_J(x) \le T d_x/2 \le 2T(d_x - 3).$$

The last inequality follows from $d_x \ge 4$ and $T \ge 0$. Summing it bounds the third class by $2TE_C$. Combining the three bounds and Lemma 6.1 yields

$$\begin{aligned} I &\le 2m + \tfrac{3}{2}Tb + 2TE_C \\ &\le 2m + 6T\Delta_J + 2TE_C \\ &\le 2m + 6T(\Delta_J + E_C) = 2m + 6T \operatorname{def}(W), \end{aligned}$$

where the last equality is (*). The $12T$ version follows from $\operatorname{def}(W) \ge 0$. □

7 Charging hyperedges missed by a dominating set

Fix an arbitrary dominating set D, and a domination assignment

$$f : V(G) \to D, \qquad f(w) \in N[w], \qquad f(d) = d \quad (d \in D).$$

This choice is only for analysis; the algorithm does not know D or f. Let m_1, m_2 count the singleton and pair edges of $\mathcal{H}$ missed by D. Let $\mathcal{P}$ be the graph of the m_2 missed pairs; its maximum degree is at most T.

For $0 < t \le 1/14$, recall

$$W_t = S \cup \{v \in N[S] \setminus S : y_v \ge t\}, \qquad S_0 = \{s \in S : y_s = 0\},$$
$$D_b = \{d \in D : y_d < 1/14\} \cup (D \cap U), \qquad D_g = D \setminus D_b.$$

The geometric arguments use

$$D_g \subseteq W_t, \qquad y_d > 0 \quad (d \in D_g), \tag{44}$$

and

$$w \in X(e),\ e \in \mathcal{H} \quad \Longrightarrow \quad y_w = 0. \tag{45}$$

Indeed, a member of D_g has weight at least $1/14$, hence belongs to R, and does not belong to $U = R \setminus N[S]$. It therefore lies in $N[S]$ and in W_t. The second fact holds because $X(e) \subseteq X = V(G) \setminus R$ and the weights are zero outside R.

Every retained hyperedge meets S. If all endpoint rows of a hyperedge were slack, its z_e coordinate could be increased, contrary to maximality. If the redundant cap $z_e \le 1$ is attained, its unit rows are already tight. A tight row belongs to S, because $y_v \le 1/7$. In particular, every retained singleton center is in S.

Lemma 7.1 (Singleton charge)**.** *For every $0 < t \le 1/14$,*

$$(T-9)m_1 \le 2|S_0| + 2|D_b| + \mathrm{def}(W_t). \tag{46}$$

Proof. Let A be the missed singleton centers. Then $|A| = m_1$, $A \cap D = \varnothing$, and $A \subseteq S \subseteq W_t$. Each $\{a\}$ is T-hard. The set $f(N(a)) \subseteq D$ dominates $N(a)$, since each $w \in N(a)$ has its assigned dominator in this set. It avoids a, so hardness gives $|f(N(a))| \ge T+1$. Consequently

$$P = \{(a,d) : a \in A,\ d \in f(N(a))\} \quad \text{satisfies} \quad |P| \ge (T+1)m_1.$$

For each distinct pair $(a,d) \in P$, choose one witness $w = w_{a,d} \in N(a)$ with $f(w) = d$. The map

$$(a,d) \longmapsto (a, w_{a,d}) \tag{47}$$

is injective: the image recovers a and w, hence $d = f(w)$.

Partition P in the following order:

$$P_1 = \{(a,d) \in P : w_{a,d} \in A\},$$
$$P_2 = \{(a,d) \in P : w_{a,d} \notin A,\ w_{a,d} \in S_0\},$$
$$P_3 = \{(a,d) \in P : w_{a,d} \notin A \cup S_0,\ d \in D_b\},$$
$$P_4 = \{(a,d) \in P : w_{a,d} \notin A \cup S_0,\ d \in D_g\}.$$

These classes are disjoint and exhaustive. Every chosen witness has weight zero, since it belongs to $N(a) = X(\{a\})$ and (45) applies. A witness outside $A \cup S_0$ is therefore outside S and, as $t > 0$, outside W_t.

For P_1, (47) maps into the oriented edges of $G[A]$, so

$$|P_1| \le 2e_G(A) \le 6m_1.$$

For P_2, it maps into edges between the disjoint sets A and $S_0 \setminus A$, with one orientation per edge. Thus

$$|P_2| \le e_G(A, S_0 \setminus A) \le 2(|A| + |S_0 \setminus A|) \le 2(m_1 + |S_0|).$$

For P_3, Lemma 2.4, with endpoint sets A, D_b, applies because each witness avoids A. It gives

$$|P_3| \le 2(m_1 + |D_b|).$$

A witness already in D_b is allowed: it is fixed as its own assigned center and is not contracted to another vertex.

For P_4, both endpoints are in W_t, while each witness is outside W_t. The edge ad is absent from G: otherwise $d \in N(a)$ would have weight zero by (45), contradicting $d \in D_g$ and (44). The new-edge assertion of Lemma 2.4 therefore gives

$$|P_4| \le \operatorname{def}(W_t).$$

All contractions for this last class can be performed together while fixing W_t and retaining its induced edges. They add the distinct previously absent edges indexed by P_4. The contraction used for P_3 need not be the same contraction.

Adding the four estimates proves

$$(T+1)m_1 \le |P| \le 10m_1 + 2|S_0| + 2|D_b| + \operatorname{def}(W_t).$$

Subtracting $10m_1$ gives (46). □

Lemma 7.2 (Pair charge)**.** *For every* $0 < t \le 1/14$,

$$(Q - 86T - 2)m_2 \le 3T|D_b| + T|S_0| + 12T \operatorname{def}(W_t). \tag{48}$$

Proof. Let A be the endpoints of the m_2 missed pairs. Then $|A| \le 2m_2$ and $A \cap D = \varnothing$. Every missed pair e is Q-hard. Its common neighborhood $X(e)$ avoids e, because the graph has no loops, and dominates itself using closed neighborhoods. Hence $|X(e)| \ge Q + 1$. The incidence set

$$\mathcal{I} = \{(e, w) : e \in E(\mathcal{P}),\ w \in X(e)\}$$

satisfies

$$|\mathcal{I}| = \sum_{e \in E(\mathcal{P})} |X(e)| \ge (Q+1)m_2 \ge Qm_2. \tag{49}$$

With $c(w)$ as in Lemma 2.5, $|\mathcal{I}| = \sum_w c(w)$. Every witness with $c(w) > 0$ has weight zero by (45).

Partition these incidences by their witness:

(i) $w \in A$;

(ii) $w \notin A$ and $w \in S_0$;

(iii) $w \notin A \cup S_0$ and $f(w) \in D_b$;

(iv) $w \notin A \cup S_0$ and $f(w) \in D_g$.

These classes are disjoint and exhaustive. Witnesses in the last two classes lie outside W_t: they have weight zero, and membership in S would put them in S_0.

By (4), the first class has size at most

$$3T|A| \le 6Tm_2. \tag{50}$$

By (5) for $A, S_0 \setminus A$, the second class has size at most

$$T(|A| + |S_0 \setminus A|) \le 2Tm_2 + T|S_0|. \tag{51}$$

To treat multiplicities in the third class, let C_b be its set of distinct witnesses. Contract the vertices of $C_b \setminus D$ to their assigned centers as in Lemma 2.3, obtaining a simple planar graph F. Witnesses already in D are left fixed as their own centers. Every vertex of A and D_b is fixed, since C_b avoids A and no vertex of D is a leaf.

For an incidence (e, w) in this class, write $e = \{a, b\}$ and $d = f(w)$. The three vertices a, b, d are distinct, because a, b are distinct members of A and $d \in D$. The edges aw, bw become ad, bd in F. Map the incidence to (e, d). For any fixed image $(\{a, b\}, d)$, its preimages are distinct members of

$$N_G(a) \cap N_G(b) \cap N_G[d].$$

Lemma 2.2 bounds their number by three, including the possible preimage $w = d$.

Every image (e, d) has both endpoints of e in $N_F(d) \cap A$. The pair graph is still the original $\mathcal{P}$, with maximum degree at most T. Applying (5) in the planar graph F to the disjoint sets A, D_b bounds the number of possible images by $T(|A| + |D_b|)$. Multiplying by the fiber bound three gives

$$|\mathcal{I}_{\text{iii}}| \le 3T(|A| + |D_b|) \le 6Tm_2 + 3T|D_b|. \tag{52}$$

Any extra edges created by contraction only enlarge the set of possible images; that enlarged set is still bounded by the planar bipartite edge count.

For the fourth class, put $W' = W_t \cup A$ and let C_g be its set of distinct witnesses. Each $w \in C_g$ lies outside W', while $d = f(w) \in D_g \subseteq W_t$. Choose any missed pair $\{a, b\}$ witnessed by w. The three vertices a, b, d are distinct and in W'. The edges wa, wb exist. Also wd exists: domination gives either $w = d$ or adjacency, and equality is impossible because $w \notin W'$ and $d \in W'$. Hence w has at least three neighbors in W'. All pair endpoints are in W'. The $12T$ form of Lemma 6.3 gives

$$|\mathcal{I}_{\text{iv}}| \le 2m_2 + 12T \operatorname{def}(W'). \tag{53}$$

This counts precisely the incidences at witnesses in C_g, since their category depends only on w, not on which pair they witness.

Adding the vertices of A retains every old induced edge on W_t. Therefore

$$\begin{aligned} \operatorname{def}(W') &= 3|W'| - e_G(W') \\ &\le 3(|W_t| + |A|) - e_G(W_t) = \operatorname{def}(W_t) + 3|A| \le \operatorname{def}(W_t) + 6m_2. \end{aligned}$$

Thus

$$|\mathcal{I}_{\text{iv}}| \le 2m_2 + 12T \operatorname{def}(W_t) + 72Tm_2. \tag{54}$$

Combining (49) with the four bounds gives

$$Qm_2 \le (86T + 2)m_2 + 3T|D_b| + T|S_0| + 12T \operatorname{def}(W_t),$$

where $86 = 6 + 2 + 6 + 72$. Rearrangement proves the claim. For $m_2 = 0$, all the incidence sets are empty and the same inequalities remain valid. □

8 The exact endpoint

We now combine the pointwise planar charges with the budget from Section 5. Recall that m_1, m_2 count the hyperedges missed by the arbitrary dominating set D.

Lemma 8.1. *For the fixed constants in (18),*

$$Z_m \le \frac{\mathcal{R}}{100}.$$

Proof. Apply the finite average (27) on $0 < t \le 1/14$ to (46), and multiply by 14. Since the deficit is nonnegative, $\langle \mathrm{def}(W_t) \rangle_{1/14} \le J_0$, and hence

$$(T-9)m_1 \le 2|S_0| + 2|D_b| + 14J_0 \le (14+70+14)\mathcal{R} = 98\mathcal{R}.$$

Here we used Lemma 5.2 and $J_0 \le \mathcal{R}$. The same operation on (48) gives

$$\begin{aligned}(Q-86T-2)m_2 &\le 3T|D_b| + T|S_0| + 168TJ_0 \\ &\le (105+7+168)T\mathcal{R} = 280T\mathcal{R}.\end{aligned}$$

Both denominators below are strictly positive. More precisely,

$$T-9 = 999991 > 98000, \qquad Q-86T-2 = 999913999998 > 280000T = 280000000000.$$

Therefore

$$\frac{98}{T-9} < \frac{1}{1000}, \qquad \frac{280T}{Q-86T-2} < \frac{1}{1000}.$$

Every $z_e \le 1$, so

$$Z_m \le m_1 + m_2 \le \left(\frac{98}{T-9} + \frac{280T}{Q-86T-2}\right)\mathcal{R} \le \frac{\mathcal{R}}{100}.$$

The last weak inequality follows from $2/1000 < 1/100$ and $\mathcal{R} \ge 0$; it also covers $\mathcal{R} = 0$ without a separate strict-inequality argument. □

Proof of Theorem 1.1. The construction of Section 4 is an effective deterministic constant-round algorithm, by Lemma 3.1 and the implementation in Section 4.3. Its output O dominates G by Lemma 4.3. For every dominating set D, (37) and Lemma 8.1 yield

$$7|D| - |O| \ge \frac{1}{2}\mathcal{R} - \frac{13}{3}Z_m \ge \left(\frac{1}{2} - \frac{13}{300}\right)\mathcal{R} = \frac{137}{300}\mathcal{R} \ge 0.$$

A minimum dominating set exists because the graph is finite and the whole vertex set is dominating. Choosing such a D proves $|O| \le 7\gamma(G)$. All estimates used relaxed planar edge bounds valid for every order and every number of components, so no exceptional small or disconnected cases need to be removed. On the empty graph the algorithm outputs the empty set and the same inequality holds. □

The number of rounds is one fixed integer, at most $4L + 20$ for the explicit finite palette specified in Section 4.3. Neither a limit over algorithms nor an input-dependent choice of an error tolerance is involved. The packing is only coordinatewise maximal, and its exact computation uses bounded row size rather than a maximum-degree bound on the input graph. The final comparison uses an integral dominating set through its domination assignment; the theorem does not assert the stronger inequality $|O| \le 7\gamma_f(G)$.

Combined with the lower bound of Hilke, Lenzen, and Suomela [5], the theorem proves that seven is both the infimum and an attained approximation factor for deterministic constant-round planar domination.

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